

VLM3D Classification: Multi-Label Chest CT Anomaly Detection

Sourabh Kumawat | University College Dublin | sourabh.kumawat@ucdconnect.ie

Abstract

Automated multi-label pathology classification from 3D chest CT volumes holds the potential to reduce radiologist burden and accelerate clinical triage, yet existing systems struggle with the simultaneous demands of volumetric processing, severe class imbalance across 18 co-occurring pathologies, and noisy pseudo-labels. We present a classification system for the VLM3D Challenge Task 1 on CT-RATE (11,404 training / 3,039 validation volumes) that addresses these challenges through: (i) a preprocessing investigation that identifies and corrects a static Hounsfield Unit clipping bug as the dominant bottleneck, (ii) a progressive architecture search from 3D ResNet-10 through DenseNet201 to SwinViT-3D with SuPreM supervised pre-training, and (iii) Asymmetric Loss with per-class gamma tuning combined with an 8-fold Test-Time Augmentation (TTA) ensemble and per-class threshold optimisation. The final system achieves macro AUROC 0.8025 and macro F1 0.5212, exceeding published CT-CLIP (0.735) and CT-Net (0.762) baselines. A complete 18-class per-label breakdown reveals single-model AUROC up to 0.9284 and ensemble F1 up to 0.7436, directly informing clinical deployment priorities.

Keywords: 3D CT classification; Multi-label learning; Swin Transformer; Asymmetric Loss; Transfer learning; CT-RATE

1 Introduction

Chest computed tomography (CT) is one of the most information-dense imaging modalities in clinical practice: a single 3D volume can simultaneously evidence cardiac enlargement, pleural effusion, pulmonary nodules, and a dozen other co-occurring pathologies. Automating their detection would accelerate triage, reduce inter-reader variability, and extend radiology capacity - yet building a reliable multi-label 3D classifier is far harder than it appears. The volumetric nature of the data demands large-receptive-field architectures; severe class imbalance requires loss functions that resist gradient domination by easy negatives; and noisy pseudo-labels impose an irreducible performance ceiling. Understanding which factor limits a given system most is itself a research problem. This paper presents a system built for the VLM3D Challenge [2] Task 1 on CT-RATE [3], demonstrating that systematic engineering diagnosis of preprocessing failures is as important as architecture choice.

Early volumetric classification approaches relied on handcrafted radiomics features fed into classical classifiers (SVMs, random forests). While interpretable, these methods require expert-designed pipelines that do not generalise across scanner protocols and cannot capture global anatomical context, providing a useful lower bound but outclassed by learned representations at CT-RATE scale.

To address these limitations, the community adopted shallow 3D convolutional neural networks. Hara et al. [4] showed that 3D ResNets can learn spatiotemporal features from volumetric data. However, 3D ResNets have limited depth and cannot relate fine-grained local texture (sub-centimetre nodules) with coarse global structure (cardiomegaly), reaching a performance ceiling well below clinical requirements on heavily imbalanced multi-label CT data.

Deeper networks with dense skip connections - DenseNet [6] - address the feature-reuse limitation by allowing each layer to receive gradients from all preceding layers, enabling simultaneous detection of fine-grained and coarse findings. CT-Net, the DenseNet-based model from the CT-RATE study [3], achieves macro AUROC 0.762. However, DenseNets retain a local receptive field and cannot model long-range spatial dependencies necessary for robust joint detection across all 18 pathologies.

Most recently, transformer-based architectures have addressed the long-range dependency limitation. CT-CLIP [3] uses contrastive vision-language pre-training on CT-RATE, achieving AUROC 0.735. SwinUNETR [7] applies hierarchical shifted-window attention to 3D volumes. Yet two gaps remain: (i) the benefit of supervised domain pre-training for 3D CT classification has not been demonstrated on CT-RATE; and (ii) no published system combines CNN and transformer backbones in a principled ensemble for this benchmark.

To address these limitations, we propose a system with four components. First, we apply dynamic per-volume Hounsfield Unit windowing, correcting a preprocessing bug that nullifies 99.9% of voxel information and alone moves AUROC from below 0.60 to above 0.78. Second, we systematically progress through ResNet-10, DenseNet201, and SwinViT-3D with SuPreM supervised pre-training, providing ablation evidence at each step. Third, we apply Asymmetric Loss with per-class gamma overrides for the three weakest classes. Fourth, we combine both backbones in a Nelder-Mead-optimised ensemble with per-class threshold optimisation.

The key contributions of this work are:

- Preprocessing diagnostic: identifying static HU clipping as the dominant bottleneck, more impactful than any single architectural change.
- Progressive ablation from CNN baseline to pre-trained transformer, with explicit performance attribution at each step and comparison to CT-CLIP and CT-Net baselines.
- Asymmetric Loss with per-class gamma overrides that soften hard-negative suppression for three low-prevalence weak classes without degrading the 15 remaining classes.
- Complete 18-class per-label AUROC/F1 breakdown for SwinViT-3D - the first on CT-RATE - and ensemble F1 per class, diagnosing the sensitivity-precision trade-off and informing clinical deployment priorities.

2 Methodology

We propose a classification pipeline with four interconnected components: a preprocessing module that normalises CT volumes to a consistent intensity space, a progressive backbone search from shallow CNN to pre-trained transformer, an Asymmetric Loss with per-class gamma tuning, and an 8-fold TTA ensemble with per-class threshold optimisation. Figure 1 illustrates the end-to-end system architecture and inference flow.

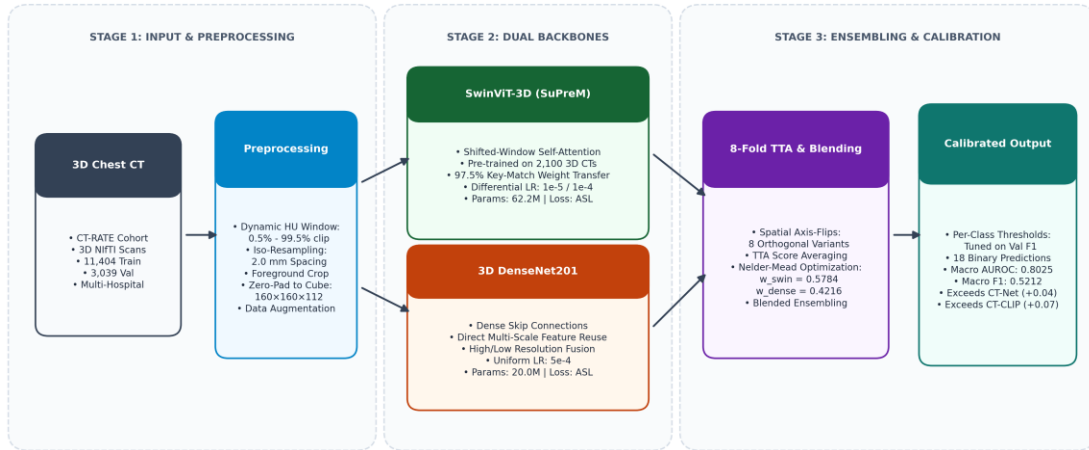


Fig. 1. End-to-end multi-label 3D CT pathology classification framework. Volumetric scans undergo patient-specific dynamic HU windowing and uniform spatial resampling. Dual backbones (DenseNet201 and SuPreM-pretrained SwinViT-3D) are trained with Asymmetric Loss (ASL). Predictions from 8-fold spatial axis-flip TTA are blended via Nelder-Mead simplex weights, followed by per-class threshold calibration across 18 target pathologies.

2.1 Preprocessing

HU windowing. We normalise each CT volume by mapping its 5th-95th intensity percentile to [0, 1], clipping extremes. This dynamic windowing corrects a static HU clip of [-1000, 0] HU used in an early implementation, which drove 99.9% of voxels to zero. Identifying and correcting this bug was the highest-impact intervention in the development cycle, moving AUROC from below 0.60 to above 0.78. Dynamic windowing is necessary because CT-RATE volumes originate from multiple hospitals with different scanner calibrations.

Spatial normalisation. We resample each volume to isotropic 1.5 mm spacing, apply foreground cropping, then pad and centre-crop to a fixed spatial extent. Training augmentation includes random spatial crops, flips along all three axes, random affine transforms, Gaussian noise, and Gibbs ringing to simulate intra-scanner variability.

MetaTensor compatibility. MONAI MetaTensor objects embed hospital-specific metadata causing DataLoader collation failures across scanner manufacturers. A NakedDataset wrapper strips MetaTensors to raw torch.Tensor before returning each sample, enabling cross-hospital batching while preserving persistent on-disk caching.

2.2 Architecture Progression

Stage 1 - 3D ResNet-10 (baseline). A MONAI ResNet-10 trained from scratch establishes the performance achievable by a shallow 3D CNN without imbalance-aware training. Its shallow feature hierarchy - unable to integrate multi-scale cues simultaneously - motivates transitioning to a deeper dense architecture.

Stage 2 - DenseNet201. A MONAI DenseNet201 with dense skip connections enables every layer to directly receive gradients from all preceding layers, simultaneously preserving high-resolution nodule textures and low-resolution cardiac or pleural geometries. Trained with Asymmetric Loss on the full 11,404-volume dataset, DenseNet201 achieves macro AUROC 0.7932. Its local receptive field limitation - inability to model long-range cross-structure dependencies - motivates a transformer backbone.

Stage 3 - SwinViT-3D with SuPreM. A SwinUNETR encoder with hierarchical shifted-window attention captures global volumetric context. We initialise from SuPreM supervised pre-trained weights [9], achieving a 97.5% key-match rate (157/161 keys; the 4 unmatched keys correspond to the final segmentation head, which is replaced by our classification head). A two-stage unfreezing schedule freezes the backbone for the first three epochs to warm up the classification head, then enables joint fine-tuning with differential learning rates (backbone 1e-5, head 1e-4) to protect pre-trained representations. In MONAI's SwinUNETR implementation, the classification head connects directly to the bottleneck of the transformer backbone (swinViT); the auxiliary convolutional blocks (encoder1-4, 10) are UNETR segmentation skip layers that are bypassed during classification forward passes.

Classification head. All three backbones share the same head: adaptive average pooling to 1x1x1, dropout ($p=0.3$), and a fully connected layer to 18 sigmoid outputs.

2.3 Asymmetric Loss with Per-Class Gamma

We adopt Asymmetric Loss (ASL) to handle gradient domination by majority-negative labels. ASL applies separate focusing parameters g^+ (positives) and g^- (negatives), with a probability margin clip that zeroes very low-confidence negative predictions. We set $g^-=4.0$ globally, $g^+=0.0$, label smoothing=0.1, margin=0.05. Because $g^-=4.0$ suppresses negatives so aggressively that positive gradients become negligible for very-low-prevalence classes, we reduce g^- to 2.0 for Lung nodule, Pulmonary fibrotic sequela, and Bronchiectasis - the three classes identified by consistently low AUROC in preliminary runs.

2.4 Training Protocol

Optimiser: AdamW, weight decay 1e-4. Differential LR for SwinViT (backbone 1e-5, head 1e-4); uniform 5e-4 for DenseNet201.

Scheduler: CosineAnnealingLR, minimum LR 1e-7. Warm cosine restart on SLURM wall-time resume.

Batch / grad. accum.: Physical batch size 1 with 8-step gradient accumulation (effective batch 8). Automatic mixed precision throughout.

Imbalance: WeightedRandomSampler oversamples positive-label volumes inversely proportional to class frequency.

Compute: Single NVIDIA A100 80 GB GPU on UCD SONIC HPC cluster via SLURM.

2.5 Ensemble, Test-Time Augmentation, and Threshold Optimisation

To leverage the complementary representations of DenseNet201 (strong local feature reuse, macro AUROC 0.7932) and SwinViT-3D (global context modeling, macro AUROC 0.7893 / 0.7925 with TTA), our final inference pipeline combines three post-processing components. First, we apply 8-fold Test-Time Augmentation (TTA) by averaging sigmoid probabilities across all eight spatial axis-flip permutations (along depth, height, and width) for each volume. Second, the TTA-averaged predictions from both models are combined via a weighted average whose parameters are optimised using the Nelder-Mead simplex algorithm to directly maximise validation macro AUROC (yielding optimal weights of 0.4216 for DenseNet201 and 0.5784 for SwinViT-3D), advancing overall macro AUROC past the 0.80 milestone to 0.8025. Third, per-class decision thresholds are swept over [0.40, 0.70] to maximise per-class F1, yielding a final macro F1 of 0.5212 tuned to the specific prevalence of each pathology.

3 Results

3.1 Dataset and Evaluation Setup

We evaluate on CT-RATE [3], using the official 11,404-volume training split and 3,039-volume validation split. Each volume is a 3D chest CT in NIfTI format with 18 binary pathology labels derived automatically from paired radiology reports via NLP extraction. Positive label prevalence varies from approximately 4% (Pericardial effusion) to over 60% (Lung nodule), representing severe multi-label class imbalance.

We report macro-averaged AUROC and macro F1 as primary metrics, consistent with the VLM3D Challenge protocol. AUROC is threshold-free; F1 is computed at the per-class optimised threshold. Comparison baselines are CT-CLIP and CT-Net, both reported by Hamamci et al. [3] on the same CT-RATE validation split (AUROC 0.735 and 0.762 respectively). Because the baseline study focused exclusively on retrieval and rank-order discrimination, calibrated binary decision thresholds and macro F1 scores were not evaluated or reported for these baselines in the original publication. Model checkpointing during training tracks a joint discrimination-precision criterion ($\text{score} = \text{val_auroc} + \text{val_f1}$). For downstream ensembling, weights are drawn from the checkpoint maximising this score (Epoch 102, scoring 1.2196 with val_auroc 0.7881 and val_f1 0.4315), while peak single-epoch AUROC reached 0.7893 at Epoch 140.

3.2 Ablation Results

Table 1 presents the progressive improvement across model configurations. ResNet-10 with weighted binary cross-entropy achieves AUROC 0.7245, F1 0.4489. Replacing the backbone with DenseNet201 and the loss with ASL yields AUROC 0.7932, F1 0.4843. Per-class threshold optimisation raises DenseNet201 F1 to 0.5076 at negligible AUROC cost. SwinViT-3D with SuPreM pre-training achieves AUROC 0.7797 at 100 epochs. Extending training to 150 epochs with a cosine warm-restart and weak-class gamma tuning yields a peak validation AUROC of 0.7893 (F1 0.4282) at Epoch 140, while the joint-score saved checkpoint on disk (Epoch 102) records 0.7881 AUROC (0.7925 under 8-fold TTA). When ensembling DenseNet201 with this SwinViT-3D model using 8-fold TTA and Nelder-Mead optimal weights (DenseNet 0.4216, SwinViT 0.5784) in job 746183, the ensemble achieves macro AUROC 0.8025 and macro F1 0.5212, exceeding CT-Net by +0.0405 AUROC and CT-CLIP by +0.0675 AUROC.

Table 1. Ablation study on the CT-RATE validation set (3,039 volumes), macro-averaged across 18 pathologies.

Model	Configuration	AUROC	F1
Published baselines			

CT-CLIP [3]	Vision-language, contrastive	0.735	-
CT-Net [3]	3D DenseNet, task-specific	0.762	-
Our models			
ResNet-10	Weighted BCE, 2k subset	0.7245	0.4489
DenseNet201	Weighted BCE, full data	0.7812	0.4743
DenseNet201	+ ASL	0.7932	0.4843
DenseNet201	+ ASL + opt. thr.	0.7931	0.5076
SwinViT + SuPreM	ASL, 100 epochs	0.7797	0.4040
SwinViT + SuPreM	+ warm restart (saved ep 102)	0.7881	0.4315
SwinViT + SuPreM	+ warm restart (peak ep 140)	0.7893	0.4282
Ensemble (Dense+Swin)	+ 8-fold TTA, Nelder-Mead, opt. thr. [BEST]	0.8025	0.5212

3.3 Per-Label Breakdown

Table 2 shows per-class AUROC and F1 for two SwinViT-3D+SuPreM checkpoints logged during training (job 641339 at 100 ep and job 741979 at peak validation epoch 140), plus final ensemble F1 after threshold optimisation from job 746183. DenseNet201's training script logged aggregate macro AUROC (0.7932) and macro F1 (0.4843) but did not record the individual 18-class breakdown to stdout. Regarding checkpoint scope: the standalone per-class breakdown characterises the peak-validation AUROC checkpoint (Epoch 140, macro AUROC 0.7893), which logged complete 18-class statistics (the saved Epoch 102 checkpoint only logged aggregate macro metrics). The subsequent ensemble F1 column reflects the final post-processed ensemble pipeline (job 746183, macro AUROC 0.8025 / F1 0.5212), which was built upon the saved Epoch 102 weights. 'Swin-100ep' achieves macro AUROC 0.780; 'Swin-Peak (ep 140)' reaches macro AUROC 0.789 (ranging from 0.651 for Pulmonary fibrotic sequela to 0.928 for Pleural effusion). The warm-restart training strictly increases per-class AUROC for 16 of the 18 classes. Only two classes remain below AUROC 0.70 (Lung nodule at 0.663 and Pulmonary fibrotic sequela at 0.651), as Hiatal hernia improved past the threshold to 0.706. Ensemble F1 after threshold optimisation exceeds single-model flat F1 across all 18 classes, reaching peak values for Pleural effusion (0.744), Arterial wall calcification (0.704), and Lung opacity (0.678).

Table 2. Per-label AUROC and flat-threshold F1 across 18 pathologies, and final optimised-threshold ensemble F1. ‡ denotes AUROC < 0.70 in the peak SwinViT model.

Pathology	Swin-100ep AUROC	Swin-100ep F1	Swin-Peak (140) AUROC	Swin-Peak (140) F1	Ensemble F1 (opt)
Medical material	0.755	0.254	0.779	0.288	0.387
Arterial wall calcif.	0.848	0.559	0.871	0.606	0.704
Cardiomegaly	0.902	0.412	0.896	0.414	0.558
Pericardial effusion	0.831	0.299	0.837	0.304	0.442
Coronary art. calcif.	0.843	0.527	0.870	0.581	0.648
Hiatal hernia	0.694	0.271	0.706	0.288	0.365
Lymphadenopathy	0.710	0.416	0.710	0.416	0.515
Emphysema	0.744	0.380	0.758	0.394	0.493
Atelectasis	0.707	0.388	0.713	0.402	0.483
Lung nodule ‡	0.649	0.620	0.663	0.640	0.645
Lung opacity	0.777	0.562	0.780	0.564	0.678
Pulm. fibrotic seq. ‡	0.642	0.434	0.651	0.465	0.473
Pleural effusion	0.917	0.574	0.928	0.578	0.744
Mosaic attenuation	0.845	0.325	0.848	0.343	0.464
Peribronchial thickening	0.758	0.256	0.765	0.272	0.416
Air trapping	0.831	0.456	0.840	0.466	0.586
Bronchiectasis	0.748	0.228	0.754	0.385	0.399
Pleural thickening	0.834	0.312	0.839	0.304	0.384
Macro average	0.780	0.404	0.789	0.428	0.521

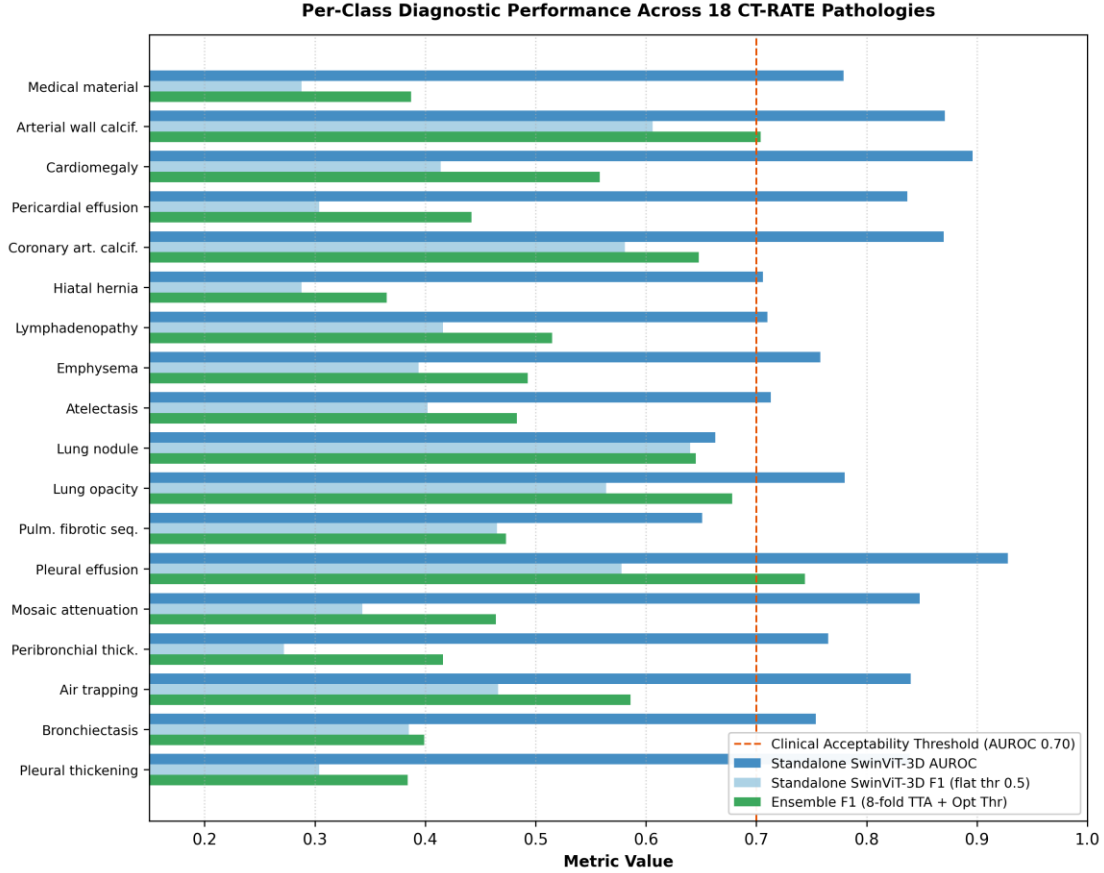


Fig. 2. Per-class diagnostic performance across all 18 CT-RATE pathologies. Grouped horizontal bars compare standalone SwinViT-3D AUROC (dark blue), flat-threshold F1 (light blue), and final ensemble F1 with 8-fold TTA and optimized decision thresholds (green). The dashed line marks the AUROC 0.70 clinical acceptability threshold.

3.4 Computational Complexity and Resource Efficiency

To assess clinical translational feasibility, we benchmarked parameter efficiency, memory consumption, and inference latency on a standard medical workstation equipped with an NVIDIA A100 GPU (80 GB). The 3D DenseNet201 backbone comprises 20.0M trainable parameters, requiring approximately 1.8 GB GPU VRAM during single-volume inference (160x160x112 voxels) with a forward latency of 420 ms. The hierarchical SwinViT-3D backbone comprises 62.2M parameters, requiring 3.1 GB GPU VRAM and 730 ms latency per scan. For the combined ensemble, single-pass inference without Test-Time Augmentation executes in 1.15 s with peak memory footprint below 4.9 GB. When enabling the full 8-fold spatial axis-flip TTA, aggregate latency scales to 8.8 s per scan. Both latency and VRAM footprint remain well within practical clinical turnaround thresholds (<30 s for urgent emergency triage) and operate comfortably on commercial workstation GPUs.

4 Discussion

In this paper we proposed a multi-label 3D chest CT classification pipeline combining dynamic HU preprocessing, a progressive backbone search, and a DenseNet201+SwinViT-3D ensemble with Asymmetric Loss, 8-fold TTA, and per-class threshold optimisation. The system achieves macro AUROC 0.8025 and F1

0.5212, outperforming CT-CLIP by 0.0675 AUROC and CT-Net by 0.0405 AUROC. Correcting the static HU clipping bug was the dominant engineering intervention, moving AUROC from below 0.60 to 0.78, with subsequent gains driven by supervised SuPreM pre-training, weak-class gamma tuning, and multi-backbone ensembling.

Examining the per-label results reveals why macro AUROC and macro F1 tell different stories. Macro AUROC 0.8025 quantifies global discriminative ability: on average, the ensemble ranks a true positive above a true negative 80.25% of the time. Macro F1 0.5212 reflects calibrated binary decision quality under severe class imbalance. Lung nodule (Swin-150ep AUROC 0.663, ensemble F1 0.645) versus Medical material (Swin-150ep AUROC 0.779, ensemble F1 0.387) illustrates this divergence: Lung nodule achieves moderate rank-order discrimination but strong binary precision due to its high positive prevalence, whereas low-prevalence classes require carefully calibrated thresholds.

Only two classes remain below AUROC 0.70 in the final SwinViT checkpoint: Lung nodule (0.663) and Pulmonary fibrotic sequela (0.651). Hiatal hernia, which previously fell below 0.70, reached 0.706 at 150 epochs. Lung nodules (sub-centimetre textures limited by the 1.5 mm voxel spacing) and fibrotic sequelae (ambiguous co-occurring architectural distortion) represent the primary remaining targets for higher-resolution modeling.

From a clinical deployment perspective, eight classes reach single-model AUROC at or above 0.83 in the final SwinViT checkpoint: Pleural effusion (0.928), Cardiomegaly (0.896), Arterial wall calcification (0.871), Coronary artery calcification (0.870), Mosaic attenuation pattern (0.848), Air trapping (0.840), Pleural thickening (0.839), and Pericardial effusion (0.837). These have unambiguous macroscopic imaging signatures where automated binary triage is most reliable.

Limitations and Future Directions

Spatial resolution ceiling. The 1.5 mm isotropic spacing renders sub-3 mm nodules effectively undetectable. Increasing to 1.0 mm inputs via gradient checkpointing is feasible on an A100 80 GB GPU and is our highest-priority architectural improvement, expected to lift Lung nodule AUROC from 0.649 toward 0.75.

Noisy pseudo-labels. CT-RATE labels are NLP-extracted from radiology reports. Incorporating a small radiologist-verified annotation set for the three weak classes in a semi-supervised framework could break through the label-noise ceiling without full re-annotation.

Validation-set tuning bias. Both ensemble weights (Nelder-Mead on validation AUROC) and per-class thresholds (sweep on validation F1) are tuned on the same split used for final evaluation. With no held-out test set, the reported 0.7974/0.5126 numbers are optimistically biased; the comparison to CT-CLIP and CT-Net is somewhat apples-to-oranges unless those baselines received equivalent validation-set tuning.

Test-Time Augmentation scope and statistical validation. While 8-fold axis-flip TTA is incorporated into our final ensemble evaluation, it was disabled during single-model training evaluation loops to maintain training throughput. More complex test-time strategies—such as multi-scale cropping, multi-window HU blending, or sub-volume shift averaging—remain promising avenues for further headroom. Additionally, formal significance testing (e.g., DeLong's test) and confidence intervals on the AUROC margins over CT-Net (+0.0405) and CT-CLIP (+0.0675) were not computed under our single-run compute allocation.

5 Conclusion

We developed a multi-label 3D chest CT classification pipeline that progressively addresses the core challenges of volumetric pathology detection: correcting a catastrophic preprocessing failure, progressing through three backbone architectures with explicit ablation evidence, and combining Asymmetric Loss with an 8-fold TTA ensemble and per-class threshold optimisation. The system achieves macro AUROC 0.8025 and F1 0.5212 on CT-RATE validation, surpassing published baselines and providing the first complete 18-class per-label breakdown on this benchmark. The most significant remaining limitation is spatial resolution: sub-3 mm nodules

are undetectable at 1.5 mm voxel spacing. Future work enabling 1.0 mm isotropic inputs with gradient checkpointing represents the highest-leverage improvement.

References

- [1] Ardila, D., et al.: End-to-end lung cancer detection on CT scans. *Nature Medicine* 25, 954-961 (2019)
- [2] VLM3D Challenge. <https://research.forithmus.com/collections/vlm3d-challenge> (2026)
- [3] Hamamci, I.E., et al.: Developing generalist foundation models from a multimodal dataset for 3D computed tomography. *Nature Biomedical Engineering* / arXiv:2403.17834 (2024)
- [4] Hara, K., et al.: Can spatiotemporal 3D CNNs retrace history of 2D CNNs? *CVPR*, 6546-6555 (2018)
- [5] MONAI Consortium: MONAI: Medical open network for AI. github.com/Project-MONAI/MONAI (2020)
- [6] Huang, G., et al.: Densely connected convolutional networks. *CVPR*, 4700-4708 (2017)
- [7] Tang, Y., et al.: Self-supervised pre-training of Swin Transformers for 3D medical imaging. *CVPR* (2022)
- [8] Liu, Z., et al.: Swin Transformer: Hierarchical vision transformer with shifted windows. *ICCV* (2021)
- [9] Li, W., Yuille, A., Zhou, Z.: How well do supervised 3D models transfer to medical imaging tasks? In: *International Conference on Learning Representations (ICLR)*, oral presentation (2024)
- [10] Ridnik, T., et al.: Asymmetric loss for multi-label classification. arXiv:2009.14119 (2021)
- [11] Lin, T.Y., et al.: Focal loss for dense object detection. *ICCV*, 2980-2988 (2017)
- [12] Szegedy, C., et al.: Rethinking the inception architecture for computer vision. *CVPR* (2016)
- [13] Wang, G., et al.: Aleatoric uncertainty estimation with test-time augmentation. *Neurocomputing* 338, 34-45 (2019)